

PARAS BANSAL

🌐 Portfolio | ✉ parasb736@gmail.com | 📞 +91-7889099620 | 🔗 LinkedIn | 🏠 GitHub

Education

Indian Institute of Information Technology, Allahabad
B.Tech in Information Technology; CGPA: 9.00

Aug 2024 – May 2028
Prayagraj, Uttar Pradesh

Golden Earth Global School, Sangrur
Class 12 (CBSE); Percentage: 94.6%

Apr 2023 – Mar 2024
Sangrur, Punjab

Golden Earth Global School, Sangrur
Class 10 (CBSE); Percentage: 97%

Apr 2021 – Mar 2022
Sangrur, Punjab

Competitive Programming

CodeChef (**4-Star**, Rating 1820) | Codeforces (**Expert**, Rating 1710) | LeetCode (**Knight**, Rating 1914)

Key Projects

Pseudo2CPP | C++, NLP, Compiler Design, DSA | 🏠 GitHub

Feb 2026 – Mar 2026

- Implemented an **NLP-driven C++ transpiler** converting English algorithmic queries into optimized **DSA implementations** using **50+ reusable templates**.
- Designed a **semantic keyword engine** analyzing **100+ tokens** to classify algorithms with **90%+ accuracy**.
- Built modular **parsing, semantic analysis, and code generation pipelines** enabling automated competitive programming boilerplate with **O(1)** template retrieval.

Energy-Aware Beam-ACO Routing | C++, Metaheuristics, Optimization | 🏠 GitHub

Oct 2025 – Dec 2025

- Implemented a hybrid **Ant Colony Optimization (ACO) + Beam Search** routing algorithm in C++, reducing path-planning cost by **15%**.
- Integrated **Beam Search top-K pruning** to limit exploration space, improving convergence speed by **40%**.
- Designed a physics-based **energy cost model** using velocity and drag, improving routing efficiency by **15%**.

Samadhaan | MERN Stack, REST APIs, MongoDB | 🏠 GitHub | 📄

Jun 2025 – Aug 2025

- Developed a scalable **MERN-based grievance management platform** supporting **12+ civic complaint categories** with real-time status tracking.
- Built and documented **15+ RESTful APIs** using **Node.js and Express.js** with **JWT authentication** and role-based access control.
- Optimized backend performance using **MongoDB indexing**, reducing API latency by **20%** and improving UI load time by **25%**.

Technical Skills

Languages: C++ (STL), C, Java, JavaScript (ES6+), Python (Intermediate), HTML5, CSS3, SQL

Technologies/Frameworks: Git, GitHub, Linux, React.js, Node.js, Tailwind CSS, NumPy, MongoDB, MySQL

Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Theory of Computation

Achievements

- Global Top 5% (Expert) on Codeforces** with rating **1710**; ranked **830 / 91,904+** active users in India with best contest rank **451** in Codeforces Round 1066 (Div.1 & 2).
- Achieved **Global Rank 93** in CodeChef Starters 224 among thousands of participants worldwide.
- Secured **Global Rank 38** in LeetCode Weekly Contest 484.
- Solved **1100+** problems across diverse **Data Structures & Algorithms** topics on Codeforces, CodeChef, and LeetCode.
- JEE Main 2024: **AIR 3341 (Top 0.2%** among 1.4M candidates); JEE Advanced 2024: **AIR 9232**.

Leadership & Volunteering

Head Boy | *Golden Earth Global School*

May 2021 - April 2022

- Led a student **council of 35 members** to organize **10+ school events**, improving school-wide communication.

Student Mentor | *Prayaas (Social Service)*

2024 – Present

- Mentored **100+ underprivileged students**; developed resources rated **90% Excellent**.